

SANTOSH KARE

AI/ML Engineer • Full Stack Developer • Java Developer • Computer Vision Enthusiast
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EDUCATION

B.Tech in CSE (AI/ML & IoT) — GLA University, Mathura

Expected: June 2027

Bachelor of Technology | Specialization: Artificial Intelligence, Machine Learning & IoT

Intermediate (Class XII) — Shri B.M. Patil PU College, Bijapur, Karnataka

May 2021

INTERNSHIP & TRAINING

AI/ML Intern | Inveneron Technologies & Business Solutions LLP

Jun 2024 – Aug 2024

- Developed an Automated Blood Group Detection prototype using CNN on fingerprint image data, eliminating the need for invasive blood sampling.
- Applied image preprocessing (OpenCV, NumPy) to extract fingerprint features and trained classification models achieving high accuracy.
- Deployed the solution with a dual-interface: desktop GUI via Tkinter and web interface via Flask.

Full Stack Development Trainee | JOVAC – Smart Full Stack Program

2024

- Completed intensive full-stack training covering React, Node.js, JavaScript, HTML/CSS, and modern web technologies.
- Built InstaChat — a real-time chat application using full-stack technologies.
- Developed Avatar Generator (React), Quiz App (vanilla JS), and Drum Instrument (HTML/CSS/JS) as capstone projects.

Java Web Development Trainee | Self-directed Learning & Online Courses

Jan 2025 – Mar 2025

- Completed structured learning in Core Java, OOP principles, JDBC, and Spring Boot framework.
- Built RESTful APIs with Spring Boot connected to MySQL databases using Spring Data JPA.
- Practised frontend integration with React.js and styled UI components using HTML5 & CSS3.

PROJECTS

Automated Blood Group Detection via Fingerprint Analysis

Jun 2024 – Aug 2024

Python • TensorFlow • OpenCV • Flask • scikit-learn | Individual Project

- Built an end-to-end ML system predicting ABO blood group from fingerprint ridge patterns — applicable in healthcare and emergency contexts.
- Implemented image processing pipeline using OpenCV, NumPy, and skimage for feature extraction from fingerprint datasets.
- Trained CNN and classification models; designed dual-interface (Tkinter desktop + Flask web app).

CyberCam – Surveillance & Security Application

Individual Project | [GitHub](#)

Python • OpenCV • Tkinter

- Developed a real-time surveillance app supporting motion detection, intrusion identification, and suspicious activity recording.
- Implemented noise filtering and object tracking using OpenCV; integrated face identification with entry-exit logging.
- Built a Tkinter GUI for live video monitoring with alert capabilities.

Smart Traffic Stress & Accident Risk Prediction System

Smart Cities | IoT | Edge AI

Python • ML • Edge AI • GenAI • RAG

- Designed a predictive traffic safety system using edge cameras and acoustic sensors to measure traffic density, speed variance, and erratic motion.
- Built ML models to predict accident probability at intersections; deployed Agentic AI for dynamic traffic signal control.
- Integrated GenAI (RAG-based) reporting for long-term risk analysis; system auto-adjusts traffic flow during adverse weather conditions.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript
Frameworks & Libraries:	Spring Boot, Spring Data JPA, TensorFlow, scikit-learn, OpenCV, Flask, React, NumPy, skimage
Web Technologies:	HTML5, CSS3, JavaScript, React.js, Node.js
Databases & Tools:	MySQL, JDBC, MongoDB, Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Power BI, Tkinter
AI/ML Domains:	Computer Vision, Classification, Image Processing, Edge AI, Agentic AI, RAG

CERTIFICATIONS & ACHIEVEMENTS

- Power BI Desktop for Business Intelligence – Certified 2024
- Data Analyst Micro Experience – SkillCred, Jul 2024
- AI/ML Internship Certificate – Inveneron Technologies & Business Solutions LLP, Aug 2024
- Hack-4-Mini 2.0 National Hackathon – Certificate of Participation (Codeoholics & GeeksforGeeks), May 2025